

O & Soergel, lec 4: Overview IV

1) Crash course on equivariant $\mathcal{D}$ -modules

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1.1) Basics

Let Y be a smooth variety & F is an algebraic group acting on Y . Two situations we'll care about:

1) $Y = G/B$, $F = N$ (or $Y = G$, $F = N \times B$)

2) $Y = G/B$, $F = B$ (or $Y = G$, $F = B \times B$).

To $F \curvearrowright Y$ we assign the (bounded) derived category $\mathcal{D}^b(Y/F)$ of F -equivariant $\mathcal{D}$ -modules on Y . It has a t -structure whose heart is the abelian category of F -equivariant $\mathcal{D}$ -modules on Y , $\mathcal{D}_Y\text{-mod}^F$ (introduced in Sec 2.1 of Lec 3 in the case when Y was affine). In some cases ($F = \{1\}$ or Case 1) $\mathcal{D}^b(Y/F) \cong \mathcal{D}^b(\mathcal{D}_Y\text{-mod}^F)$ but in general (e.g. $F = \mathbb{G}_m$, $Y = \text{pt}$) this is not the case.

1.2) Classification of irreducible objects

Our goal is to classify irreducible objects in the abelian category $\mathcal{D}_Y\text{-mod}^F$ under the following

(*) F has fin. many orbits on Y : $Y = \bigsqcup_{i=1}^k Y_i$, $Y_i = F/F_i$

Let $F_i^\circ \subset F_i$ be the connected component of $1 \in F_i \leadsto$ finite group F_i/F_i° .

Fact: The irreducible objects in $\mathcal{D}_Y\text{-mod}^F$ are in bijection with $\bigsqcup_{i=1}^k \text{Irrep}(F_i/F_i^\circ)$, $(i, \zeta_i) \mapsto \text{IC}(Y_i, \zeta_i)$

Examples: 1) If $\overline{Y_i} = Y$, $\zeta_i = \text{triv}$, then $\text{IC}(Y_i, \text{triv}) = \mathcal{O}_Y$
2) Let $Y = G/B$ & $F = N$ or B . We have Bruhat decomposition: $G = \bigsqcup_{w \in W} NwB$, where $w \in W = N_G(H)/H$ is a lift of w to $N_G(H)$, so both cases satisfy (*). Let $N_w := \{n \in N \mid nwB = wB\}$, it's unipotent $\Rightarrow N_w^\circ = N_w \Rightarrow \text{Irr}(\mathcal{D}_{G/B}\text{-mod}^N) \xrightarrow{\sim} W$ (as we should have thx to the equivalence w. $\mathcal{O}^{\text{pr}}$ from Sec 2.2 in Lec 3). And if $F = B$, then the stabilizers are $H \ltimes N_w$, also connected.

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1.3) Construction of irreducible objects.

We need tools for understanding the objects $IC(Y_i, \text{triv})$. Inside $\mathcal{D}_Y\text{-mod}^F$ we can consider the full subcategory of "holonomic modules". E.g. $\mathcal{O}_Y$ is holonomic. If Y satisfies (*), then $F \in \mathcal{D}_Y\text{-mod}^F$ is holonomic $\Leftrightarrow F$ has finite length. In general, let $\mathcal{D}_{\text{hol}}^b(Y/F)$ be the full subcategory of complexes with holonomic cohomology.

An advantage of $\mathcal{D}_{\text{hol}}^b$ is that it has 6 functor formalism. E.g. $\forall$ F -equivariant $f: Y_1 \rightarrow Y_2$ we have:

- 1) The pullback functor $f^*: \mathcal{D}_{\text{hol}}^b(Y_2) \rightarrow \mathcal{D}_{\text{hol}}^b(Y_1)$ (with usual functoriality: $(fg)^* = g^* f^*$). We have $f^* \mathcal{O}_{Y_1} = \mathcal{O}_{Y_2}$.
- 2) Its right adjoint, pushforward functor f_*

Assume (*) in Sec 1.2. Consider the inclusion $\iota_i: Y_i \hookrightarrow Y$ (locally closed). Then:

a) $\iota_{i,*} \mathcal{O}_{Y_i} \in \mathcal{D}_{\text{hol}}^b(Y/F)$ is in cohom. deg ≥ 0

$$b) \quad IC(Y_i, \text{triv}) \hookrightarrow H^0(\iota_{i,*} \mathcal{O}_{Y_i}) \quad (1)$$

it's supported (as a quasi-coherent sheaf) on $\overline{Y_i}$;

3] coker of (1) & $H^j(\iota_{i,*} \mathcal{O}_{Y_i})$ are supported on $\overline{Y_i} \setminus Y_i \ \forall j > 0$.

c) assume $\bar{Y}_i$ is smooth & $\bar{\iota}_i: \bar{Y}_i \hookrightarrow Y$ denote the inclusion. Then $IC(Y_i, \text{triv}) = \bar{\iota}_{i,*} \mathcal{O}_{\bar{Y}_i}$.

We will need the following result, equivariant case of the Decomposition theorem of Beilinson-Bernstein-Deligne-Gabber.

Thm: Let $\varphi: \tilde{Y}_i \rightarrow \bar{Y}_i$ be F -equiv't projective morphism & $\tilde{Y}_i$ is smooth. Then $\varphi_* \mathcal{O}_{\tilde{Y}_i} \cong \bigoplus_{Y_j \subset Y_i, \delta_j} IC(Y_j, \delta_j)[?]^{\oplus ?}$

Moreover if φ is an isomorphism over Y_i , then the only term w. $Y_j = Y_i$ is $IC(Y_i, \text{triv})$ in cohomological deg 0 & with multiplicity 1.

Rem*: "Riemann-Hilbert correspondence" gives a functor from $\mathcal{D}_{\text{hol}}^b(Y/F)$ to the category of equivariant constructible sheaves $\mathcal{D}_c^b(Y/F)$. It sends $\mathcal{O}_Y$ to the rk 1 constant sheaf $\underline{\mathbb{C}}_Y$ & intertwines the functors f^*, f_* w. usual pullback & pushforwards for sheaves of vector spaces.

1.3) Equivariant cohomology.

We'll be interested in Ext's from $\mathcal{O}_Y$ in $\mathbb{D}^b(Y/F)$.

For this we need an account on the equivariant cohomology. Here $K = \mathbb{C}$, so any variety is a topological space. Let X be a topological space w. (topological) F -action. To this action we assign the **equivariant cohomology** $H_F^*(X) := H^*((E \times X)/F)$, where:

- E is contractible topological space w. topol. F -action
- $\exists$ topological principal F -bundle $E \rightarrow E/F$.

If $\exists$ topological principal F -bundle $X \rightarrow X/F$, then $(E \times X)/F \rightarrow X/F$ is a loc. trivial bundle w. contractible fibers, so $H_F^*(X) = H^*(X/F)$. So, in general, one views $H_F^*(X)$ is the "correct cohomology" of X/F . It's a graded $H_F^*(pt)$ -algebra (via pullback w.r.t. $X \rightarrow pt$)

Special cases:

1) F is unipotent so isomorphic to an affine space

$\Rightarrow$ can take $E = F$ so $H_F^*(X) = H^*(X)$.

2) $F = G_m$. Then can take $E = \mathbb{C}^\infty \setminus \{0\}$, where $\mathbb{C}^\infty = \{(z_i)_{i \geq 0} \mid i \in \mathbb{Z}_{\geq 0} \text{ \& all but fin. many } z_i = 0\}$ so

$$E/F = \mathbb{C}P^\infty \text{ \& } H_F^*(pt) = H^*(\mathbb{C}P^\infty) = \mathbb{C}[x], x \in H^2(\mathbb{C}P^\infty)$$

More generally, $H_H^*(pt)$ is canonically $\mathbb{C}[h]$.

3) $H_B^*(X) \xrightarrow{\sim} H_H^*(X)$ (exercise).

Remarks: 1) Generalizing 2): if G is a connected reductive group, then $H_G^*(pt) = \mathbb{C}[g]^G$ w. $\deg g^* = 2$. From here one can deduce that for any F , $H_F^*(pt)$ is in even degrees (and hence genuinely commutative)

2) Note that $(E \times X)/F \rightarrow E/F$ is a topologically trivial fiber bundle w. fiber X . This gives a pullback homomorphism $H_F^*(X) \rightarrow H^*(X)$ factoring through

$$H_F^*(X) \otimes_{H_F^*(pt)} \mathbb{C} \longrightarrow H^*(X) \quad (2)$$

If $H^*(X)$ is in even degrees then the spectral sequence for the cohomology of total space of a fiber bundle degenerates at E_2 and (2) is iso. This will be the case

6] for X in the next section.

1.4) Case of flag & Bott-Samelson varieties.

Now we'll give examples of computation of $H_F^*(Y)$.

I) $H^*(G/B) = H_B^*(G)$ is an $H_B^*(pt) = \mathbb{C}[\hbar]$ -algebra.

One can show it's $\mathbb{C}[\hbar]^{\text{cow}} (= \mathbb{C}[\hbar] / \mathbb{C}[\hbar] \mathbb{C}[\hbar]_+^w)$, where $\mathbb{C}[\hbar]_+^w = \{f \in \mathbb{C}[\hbar]^w \mid f(0) = 0\}$; $\mathbb{C}[\hbar], \mathbb{C}[\hbar]^{\text{cow}}$ are graded so that $\deg \hbar^* = 2$ (cf. 2) above).

II) Similarly, $H_B^*(G/B) = H_{B \times B}^*(G)$ is a $\mathbb{C}[\hbar] \otimes \mathbb{C}[\hbar]$ -algebra. It is $\mathbb{C}[\hbar] \otimes_{\mathbb{C}[\hbar]^w} \mathbb{C}[\hbar]$.

III) Recall that S denotes the set of simple reflections in W . To $\underline{w} := (s_1, \dots, s_k) \in S^k$ we assign the Bott-Samelson variety $BS_{\underline{w}}$ as follows. To $s \in S$ we assign the minimal parabolic $P_s \subset G$ containing B . Let $BS_{\underline{w}}$ be the quotient of $P_{s_1} \times \dots \times P_{s_k}$ by the following B^k -action:

$(b_1, \dots, b_k) \cdot (p_1, \dots, p_k) = (p_1 b_1^{-1}, b_1 p_2 b_2^{-1}, \dots, b_{k-1} p_k b_k^{-1})$. Denote the orbit of $(p_1, \dots, p_k)$ by $[p_1, \dots, p_k]$. We have the following maps:

$$BS_{(s_1, \dots, s_k)} \longrightarrow BS_{(s_1, \dots, s_{k-1})}, [p_1, \dots, p_k] \longrightarrow [p_1, \dots, p_{k-1}] \quad (3)$$

$$BS_{(s_1, \dots, s_k)} \longrightarrow G/B, [p_1, \dots, p_k] \mapsto p_1 \dots p_k B. \quad (4)$$

$\overline{\neq}$ (4) establishes $H_B^*(BS_{\underline{w}})$ as a $H_B^*(G/B)$ -algebra.

(3) is a $\mathbb{P}^1$ -bundle. This allows to compute $H_B^*(BS_{\underline{w}})$ inductively: it is $BS_{\underline{w}} \langle -\kappa \rangle$, the Bott-Samelson bimodule over $\mathbb{C}[y] \otimes_{\mathbb{C}[y]^w} \mathbb{C}[y]$. And $H^*(BS_{\underline{w}}) = BS_{\underline{w}} \otimes_{\mathbb{C}[y]} \mathbb{C}_0 \langle -\kappa \rangle$, a graded $H^*(G/B) = \mathbb{C}[y]^{\text{cow}}$ -module.

Remark: Here's a motivation to consider $BS_{\underline{w}}$'s: pick a reduced expression $\underline{w} = (s_1, \dots, s_e)$ for w . The morphism $\varphi_{\underline{w}}: BS_{\underline{w}} \rightarrow G/B$ from (4) has image $\overline{BwB/B}$ & is a resolution of singularities of its image.

1.5) $\text{Ext}_{\mathcal{D}^b(Y/F)}^*(\mathcal{O}_Y, ?)$

Fact: For any smooth Y w. F -action, we have a graded algebra isomorphism $\text{Ext}_{\mathcal{D}^b(Y/H)}^*(\mathcal{O}_Y, \mathcal{O}_Y) = H_F^*(Y)$. In particular, $\forall F \in \mathcal{D}_Y\text{-mod}^F$, $\text{Ext}_{\mathcal{D}^b(Y/H)}^*(\mathcal{O}_Y, F)$ is a graded right $H_F^*(Y)$ -module.

Q: Under assumption (*) from Sec 1.2, how to compute

$$\overline{8} \mid \text{Ext}_{\mathcal{D}^b(Y/F)}^*(\mathcal{O}_Y, IC(Y_i, \text{triv}))?$$

Easy case: $\bar{Y}_i$ is smooth. For $\bar{\tau}_i: \bar{Y}_i \hookrightarrow Y$ we have
 $\text{Ext}^*(\mathcal{O}_Y, \text{IC}(Y_i, \text{triv})) = [3 \text{ in Sec 1.2}] = \text{Ext}^*(\mathcal{O}_Y, \bar{\tau}_{i,*} \mathcal{O}_{\bar{Y}_i})$
 $= [\text{adjunction}] = \text{Ext}^*(\bar{\tau}_i^* \mathcal{O}_Y, \mathcal{O}_{\bar{Y}_i}) = \text{Ext}^*(\mathcal{O}_{\bar{Y}_i}, \mathcal{O}_{\bar{Y}_i}) = H_F^*(\bar{Y}_i)$
 with its natural graded $H_F^*(Y)$ -module structure

General case: Define the equivariant intersection cohomology $IH_F^*(\bar{Y}_i)$ as $\text{Ext}_{\mathbb{D}^b(Y/H)}^*(\mathcal{O}_Y, \text{IC}(Y_i, \text{triv})[\dim Y_i])$.

Corollary (of BBDG Thm in Sec 1.2) Under assumption (*), if $\tilde{Y}_i$ is an H -equivariant resolution of singularities of $\bar{Y}_i$, then we have a graded $H_F^*(Y)$ -module isomorphism:

$$H_F^*(\tilde{Y}_i)[\dim Y_i] \cong IH_F^*(\bar{Y}_i) \oplus \bigoplus_{j \mid \bar{Y}_j \neq \bar{Y}_i, \sigma_j} \text{Ext}^*(\mathcal{O}_Y, \text{IC}(Y_j, \sigma_j)[?])^{\oplus ?}$$

Example: Let $F=B$, $Y=G/B$, so $H_B^*(G/B) = \mathbb{C}[y] \otimes_{\mathbb{C}[y]^W} \mathbb{C}[y]$
 Let $Y_i = BwB/B$ ($\dim Y_i = \ell(w)$), $\tilde{Y}_i = BS_w$, $\varphi = \varphi_w$. Let $\mathcal{B}'_w := IH_B^*(\bar{Y}_i)$. Then $BS_w \cong \mathcal{B}'_w \oplus \bigoplus_{u \mid \ell(u) < \ell(w)} \mathcal{B}'_u$. Later

we'll see $\mathcal{B}'_w = \mathcal{B}_w$.